Week 2 – Input / Output & Basic Logic

■ Learning Objectives

• Learn to take user input using input().
• Convert text input to numbers using int() or float().
• Use if, elif, and else for decision making.
• Understand Boolean logic and comparison operators (==, >, <, >=, <=).

■ Key Concepts

1. User Input: Use input() to read from keyboard (always returns a string).
2. Type Conversion: Wrap input with int() or float() when you need numbers.
3. Conditionals: Use if/elif/else to make decisions based on conditions.
4. Boolean Expressions: Evaluate to True or False.

Example 1 – Reading User Input

```
# Ask for name and age
name = input("What is your name? ")
age = int(input("How old are you? "))

print(f"Hello {name}, next year you will be {age + 1} years old!")
```

Example 2 – Using if/else Statements

```
temperature = int(input("Enter the temperature in °C: "))

if temperature > 30:
    print("It's a hot day!")
elif temperature >= 20:
    print("It's a nice day.")
else:
    print("It's a bit cold today.")
```

■ Practice Problems

Practice 1 – Even or Odd Checker

Ask the user to enter a number and print whether it’s even or odd.
Hint: Use the modulus operator %.

Solution:

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("Even")
else:
    print("Odd")
```

Practice 2 – Age Group Classifier

Ask the user for their age. Print 'Child' if under 13, 'Teen' if between 13–19, else 'Adult'.

Solution:

```
age = int(input("Enter your age: "))

if age < 13:
    print("Child")
elif age <= 19:
    print("Teen")
else:
    print("Adult")
```

Practice 3 – Pass/Fail Checker

Ask for a student’s score. Print 'Pass' if score ≥ 50, otherwise 'Fail'.

Solution:

```
score = int(input("Enter your score: "))

if score >= 50:
    print("Pass")
else:
    print("Fail")
```

■ Boolean Logic Examples

```
x = 10
y = 5
print(x > y)     # True
print(x == y)    # False
print(x != y)    # True

# Combine conditions with and/or
if x > 0 and y > 0:
    print("Both are positive")
```

■ Homework Challenge

Ask the user for a number of hours worked and hourly wage, then calculate total pay. If hours > 40, pay 1.5 × hourly rate for the extra hours.

Expected Solution:

```
hours = float(input("Enter hours worked: "))
rate = float(input("Enter hourly wage: "))

if hours > 40:
    pay = 40 * rate + (hours - 40) * rate * 1.5
else:
    pay = hours * rate

print(f"Total pay: ${pay:.2f}")
```